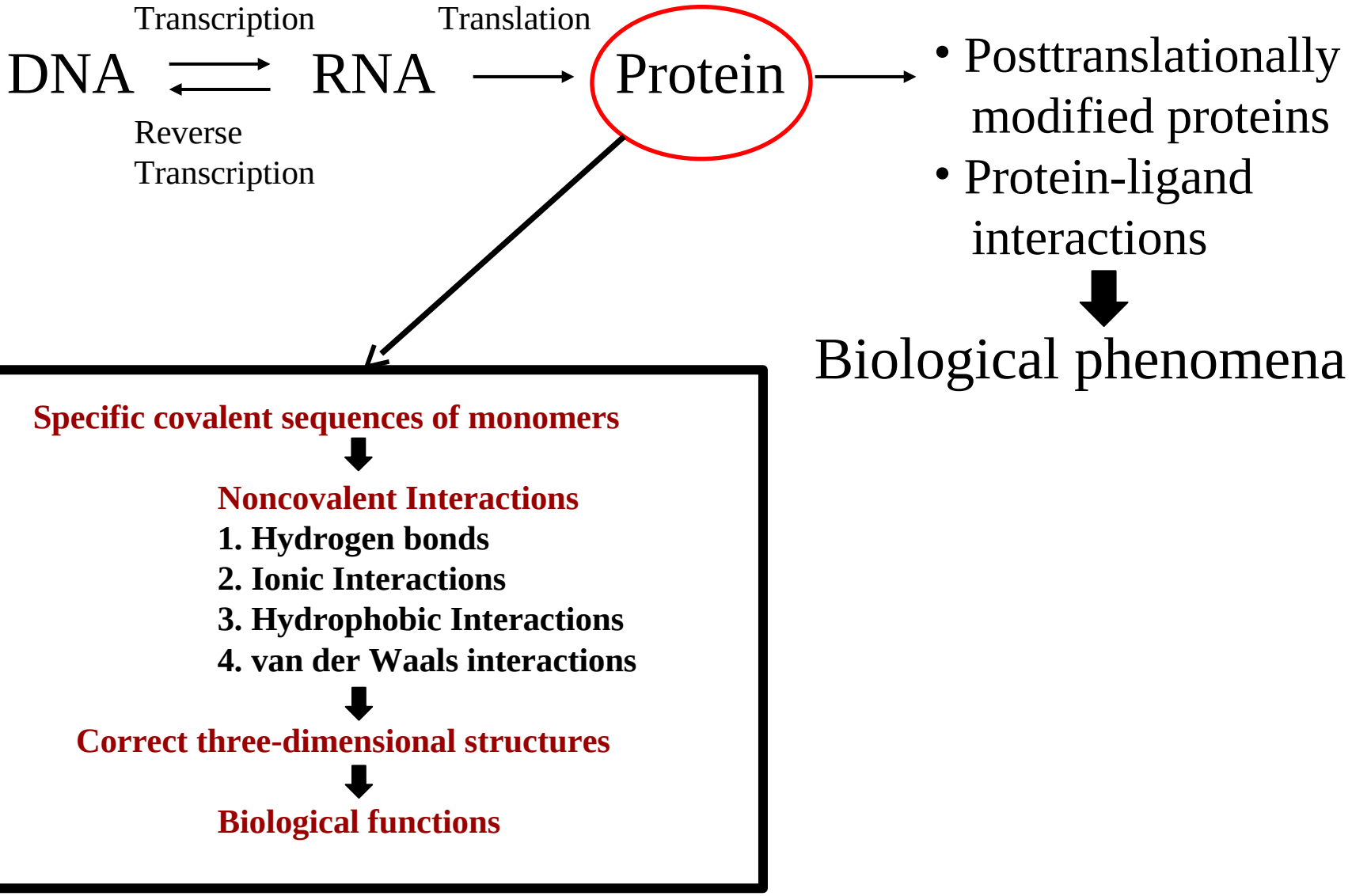
AMINO ACIDS


Transcription
Translation
DNA
RNA
Protein
Reverse
Transcription
• Posttranslationally modified proteins
• Protein-ligand interactions
Biological phenomena
Specific covalent sequences of monomers
Noncovalent Interactions
1. Hydrogen bonds
2. Ionic Interactions
3. Hydrophobic Interactions
4. van der Waals interactions
Correct three-dimensional structures
Biological functions

AMINO ACIDS

- **Amino acids are building blocks of proteins.**

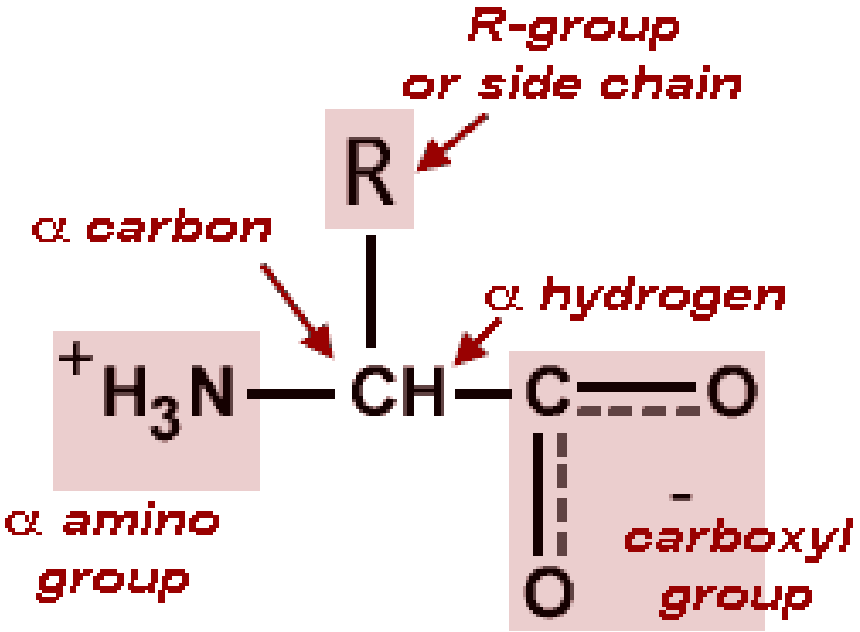
- **Proteins are composed of 20 different amino acid (encoded by standard genetic code, construct proteins in all species).**

- **Their molecules containing both amino and carboxyl groups attached to the same α-carbon (L-α-amino acids).**

- **Their chemical structure influences three dimensional structure of proteins.**

- **They are important intermediates in metabolism (porphyrins, purines, pyrimidines, creatin, urea etc).**

The basic structure of amino acids differ only in the structure of the or the side chain (R-group).

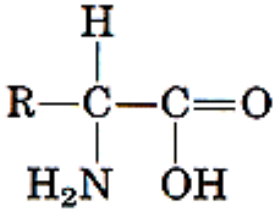


L-isomer

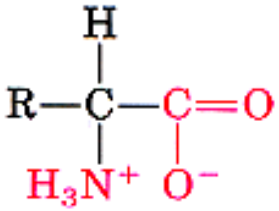
L-isomer is normally found in proteins.

Nonionic and zwitterion forms of amino acids

The zwitterion predominates at neutral pH

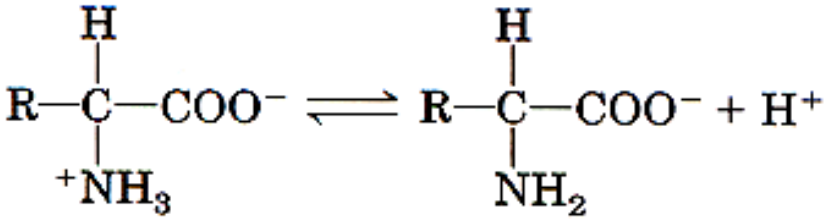


Nonionic form



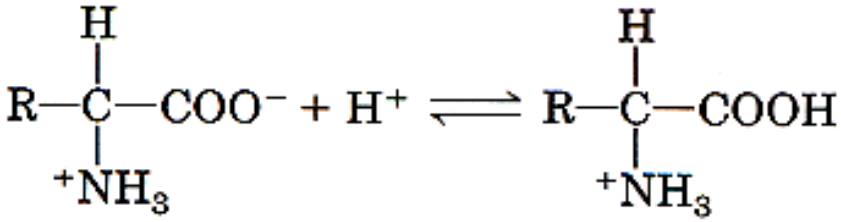
Zwitterionic form

Zwitterion = in German for „hybrid ion“



Zwitterion as acid

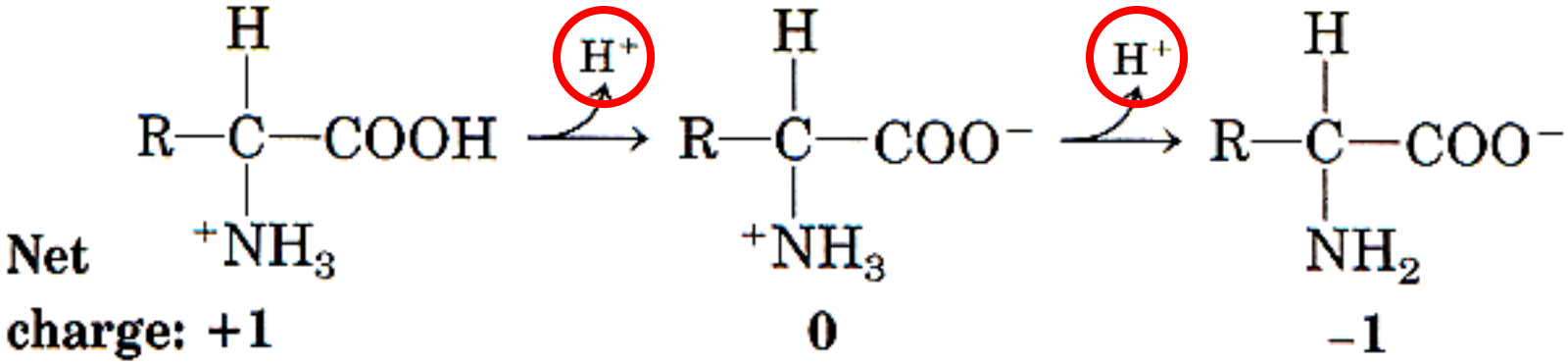
Week acid



Zwitterion as base

Week base

A simple monoamino monocarboxyl α-amino acid is a diprotic acid (can yield protons) when fully protonated



Classification of Amino Acids

Amino acids are generally divided into groups on the basis of their side chains (R groups).
The most helpful start-point is to separate amino acids into:

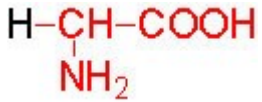
Nonpolar **Neutral polar** **Charged polar**

1. Nonpolar amino acids

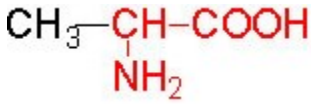
- Only carbon and hydrogen in their side chains.
- Generally *unreactive* but *hydrophobic.*
- Determining the 3-D structure of proteins (they tend to cluster on the inside of the molecule).

Nonpolar (Hydrophobic) R Groups

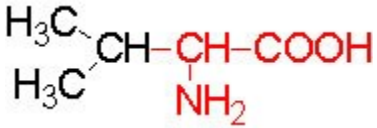
Glycine (Gly)



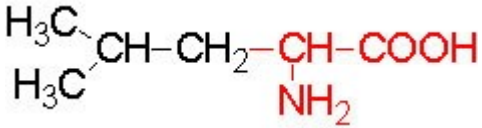
Alanine (Ala)



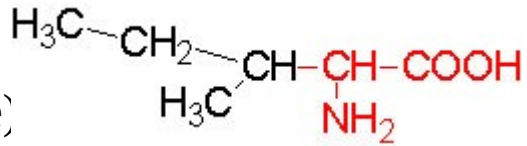
Valine (Val)



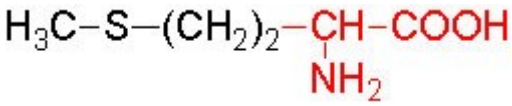
Leucine (Leu)



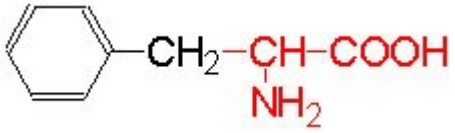
Isoleucine (Ile)



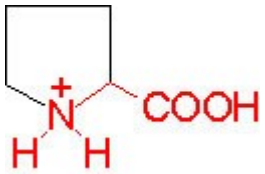
Methionine (Met)



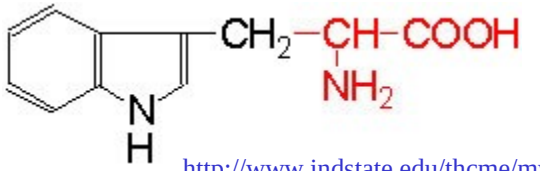
Phenylalanine (Phe)



Proline (Pro)



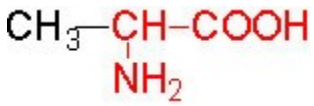
Tryptophan (Trp)



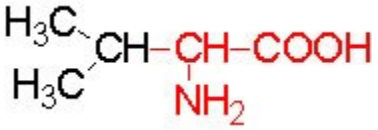
http://www.indstate.edu/thcme/mwking/amino-acids.html

The simplest amino acid is Glycine, which has a single hydrogen atom as its side chain.

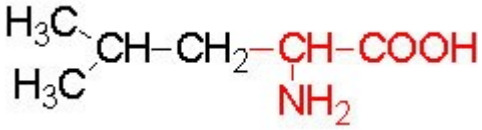
Alanine, Valine, Leucine and Isoleucine have saturated hydrocarbon R groups (i.e. they only have hydrogen and carbon linked by single covalent bonds). Leucine and Isoleucine are isomers of each other.



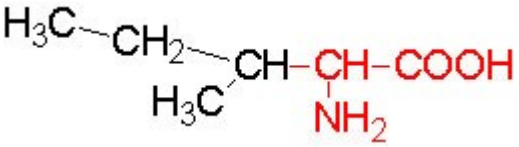
Alanine



Valine



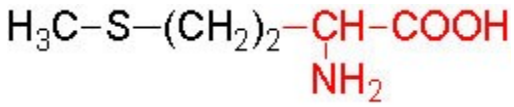
Leucine



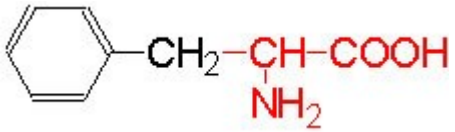
Isoleucine

- **The side chain of Methionine includes a sulfur atom but remains hydrophobic in nature.**

- **Phenylalanine is Alanine with an extra benzene (sometimes called a Phenyl) group on the end. Phenylalanine is highly hydrophobic and is found buried within globular proteins.**



Methionine

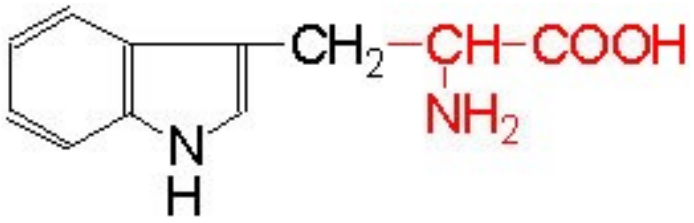


Phenylalanine

- **Tryptophan is highly hydrophobic and tends to be found immersed inside globular proteins.**

- **Structurally related to Alanine, but with a two ring (bicyclic) indole group added in place of the single aromatic ring found in Phenylalanine.**

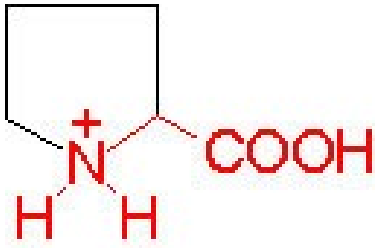
- **The presence of the nitrogen group makes Tryptophan a little less hydrophobic than Phenylalanine.**



- **Proline is unique amongst the amino acids – its side chain is bonded to the backbone nitrogen as well as to the α-carbon.**

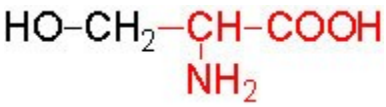
- **Because of this proline is technically an *imino* rather than an amino acid.**

- **The ring is not reactive, but it does restrict the geometry of the backbone chain in any protein where it is present.**

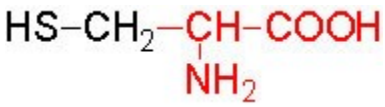


Polar (Hydrophilic) R Groups

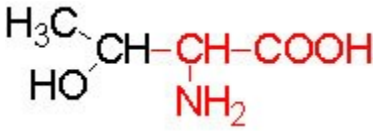
Serine (Ser)



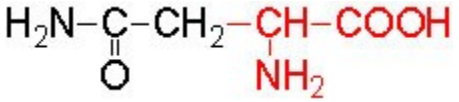
Cysteine (cys)



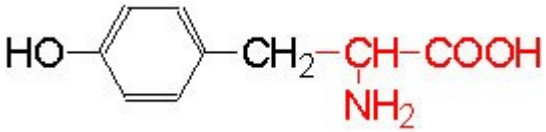
Threonine (Thr)



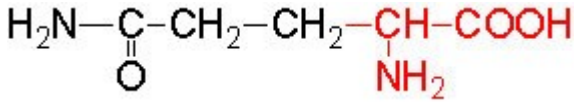
Asparagine (Asn)



Tyrosine (Tyr)

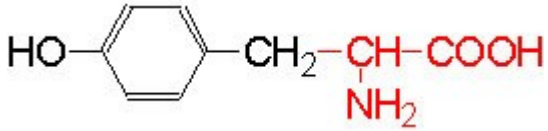


Glutamine (Gln)

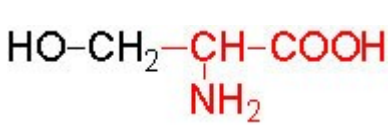


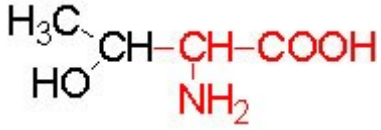
http://www.indstate.edu/thcme/mwking/amino-acids.html

- **Tyrosine is Phenylalanine with an extra hydroxyl (-OH) group attached.**
- **It is polar and very weakly acidic. Tyrosine can play an important catalytic role in the active site of some enzymes. Reversible phosphorylation of –OH group in some enzymes is important in the regulation of metabolic pathways**



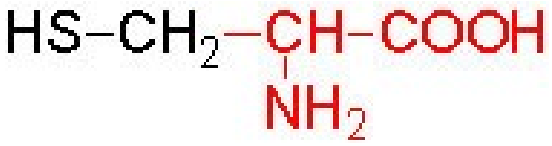
- **Serine and Threonine play important role in enzymes which regulate phosphorylation and energy metabolism.**

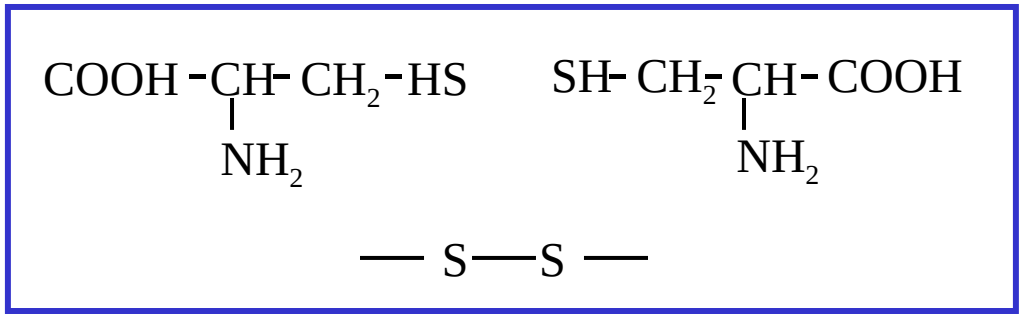




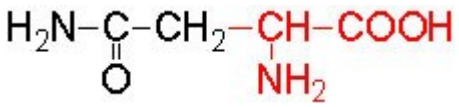
➢**Cysteine has sulfur-containing side group.The group has the potential to be more reactive.It is not very polar.**

➢**Cysteine is most important for its ability to link to another cysteine via the sulfur atoms to form a covalent disulfide bridge, important in the formation and maintenance of the tertiary (folded) structure in many proteins.**

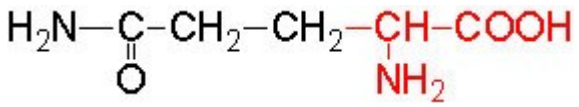




Asparagine and Glutamine are the amide derivatives of Aspartate (Aspartic acid) and Glutamate (Glutamic acid) - see below. They cannot be ionised and are therefore uncharged.



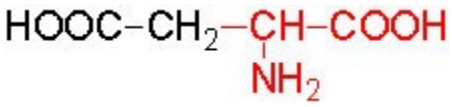
Asparagine



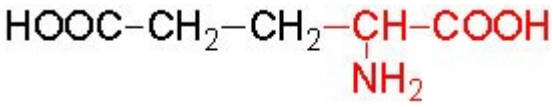
Glutamine

Negatively (Nonpolar) Charged R Groups

Aspartic acid (Asp)



Glutamic acid (Glu)

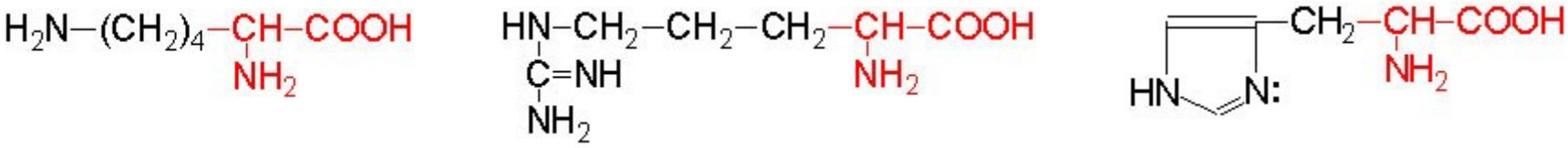


Two amino acids with negatively charged (i.e. acidic) side chains - Aspartate (Aspartic acid) and Glutamate (Glutamic acid).

These amino acids confer a negative charge on the proteins of which they are part.

Positively Charged R Groups

Lysine (Lys) Arginine (Arg) Histidine (His)



- **Lysine and Arginine both have pKs around 10.0 and are therefore always positively charged at neutral pH.**

- **With a pK of 6.5, Histidine can be uncharged or positively charged depending upon its local environment.**
- **Histidine has an important role in the catalytic mechanism of enzymes and explains why it is often found in the active site.**

Classification Based on Chemical Constitution

Small amino acids – Glycine, Alanine

Branched amino acids – Valine, Leucine, Isoleucine

Hydroxy amino acids (-OH group) **–** Serine, Threonine

Sulfur amino acids – Cysteine, Methionine

Aromatic amino acids – Phenylalanine, Tyrosine, Tryptophan

Acidic amino acids and their derivatives – Aspartate, Asparagine, Glutamate, Glutamine

Basic amino acids – Lysine, Arginine, Histidine

Imino acid - Proline

Essential Amino Acids in Humans

- Required in diet
- Humans incapable of forming requisite carbon skeleton

Arginine*	Lysine
Histidine*	Methionine
Isoleucine	Threonine
Leucine	Phenylalanine
Valine	Tryptophan

* Essential in children, not in adults

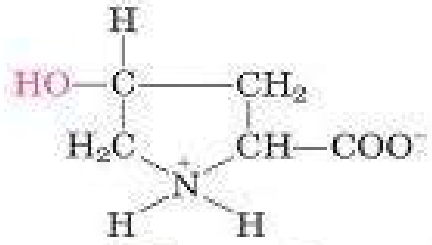
Non-Essential Amino Acids in Humans

- Not required in diet
- Can be formed from a-keto acids by transamination and subsequent reactions

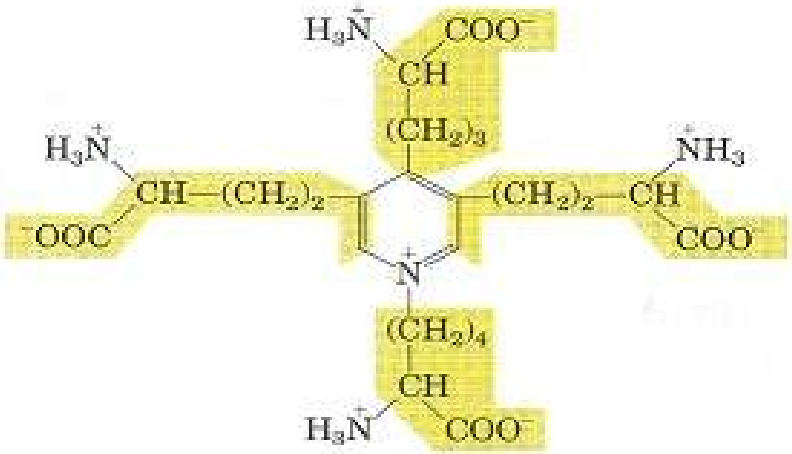
Alanine	Glycine
Asparagine	Proline
Aspartate	Serine
Glutamate	Cysteine (from Met*)
Glutamine	Tyrosine (from Phe*)

* Essential amino acids

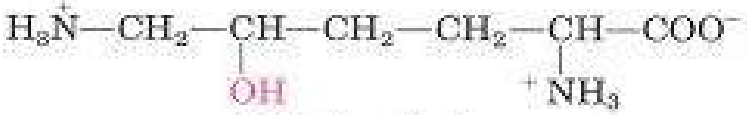
Uncommon amino acids found in proteins



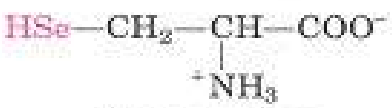
4-Hydroxyproline



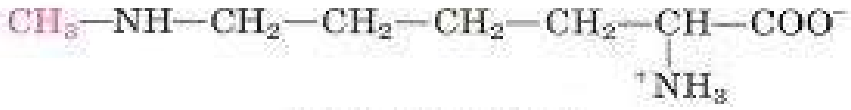
Desmosine



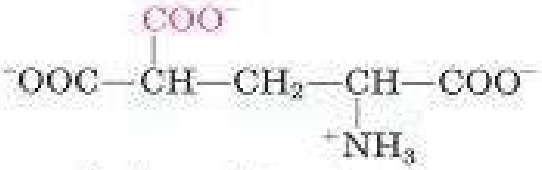
5-Hydroxylysine



Selenocysteine

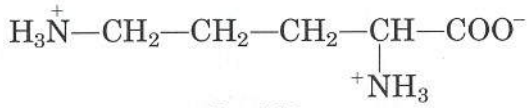


6-*N*-Methyllysine

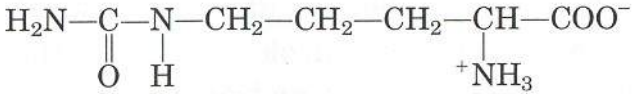


γ-Carboxyglutamate

Intermediates of biosynthesis of arginin and in urea cycle



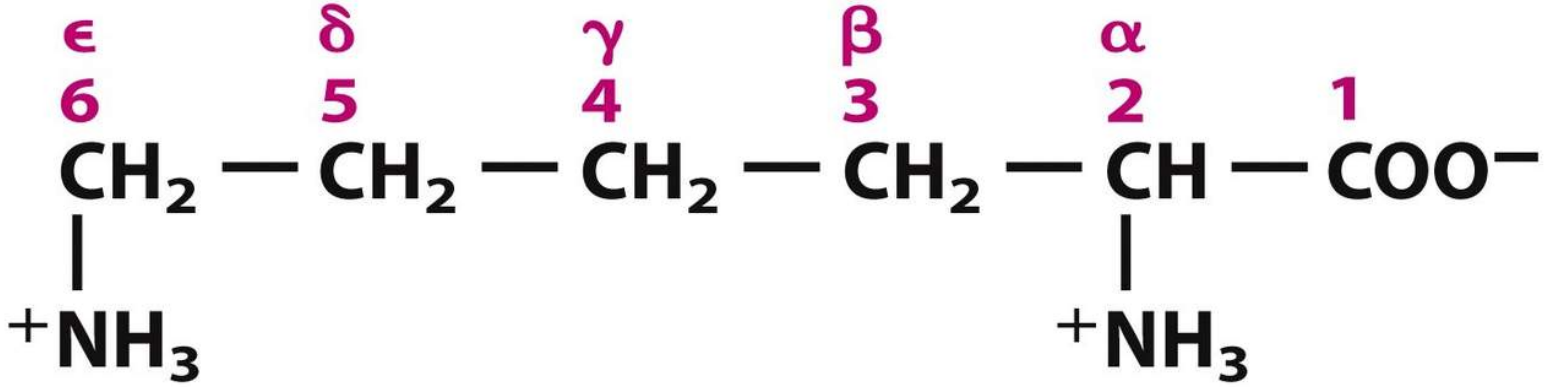
Ornithine



Citrulline

Amino Acids: Atom Naming

- Organic nomenclature: start from one end
- Biochemical designation: start from
  α-carbon and go down the R-group

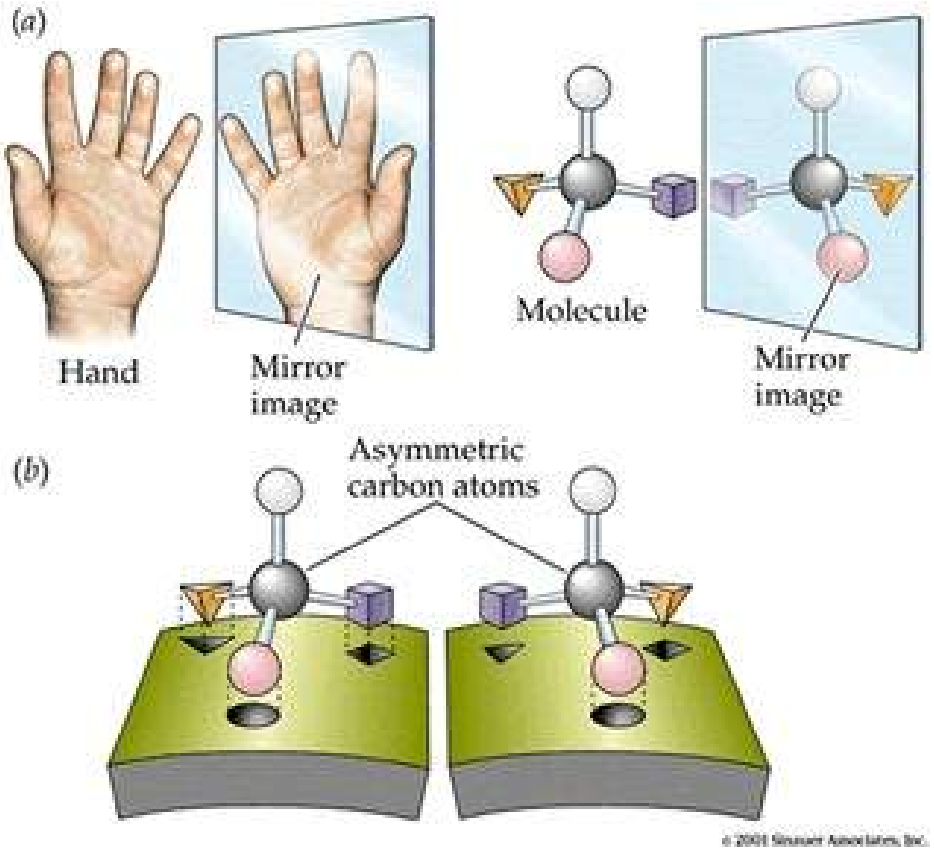


Lysine

Unnumbered 3 p74
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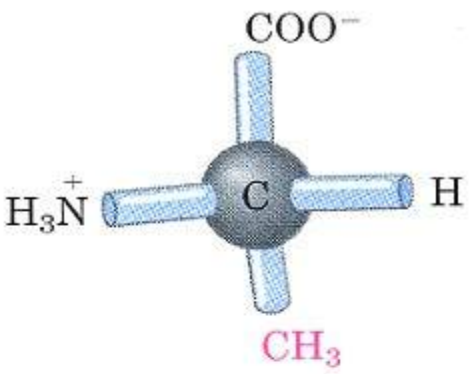
The Stereochemistry of Amino Acids

Chiral molecules existing in two forms

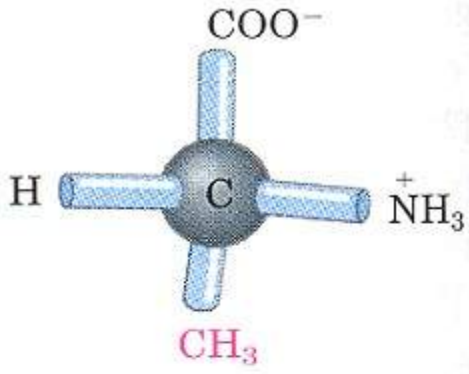


http://www.imb-jena.de/~rake/Bioinformatics_WEB/gifs/amino_acids_chiral.gif

The two stereoisomers of alanine



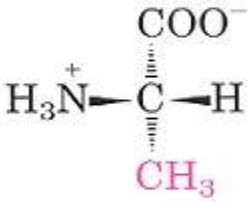
(a) L-Alanine



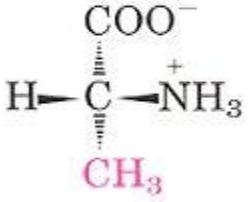
D-Alanine

α-carbon is a **chiral center**

Two stereoisomers are called **enantiomers.**

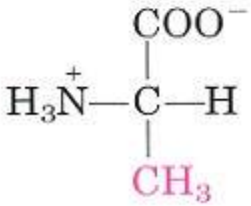


(b) L-Alanine

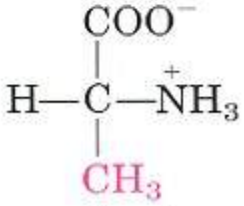


D-Alanine

The solid wedge-shaped bonds project out of the plane of paper, the dashed bonds behind it.



(c) L-Alanine



D-Alanine

The horizontal bonds project out of the plane of paper, the vertical bonds behind.

Peptides and Peptide bonds

Peptide bond in a di-peptide

“Peptides” are small condensation products of amino acids

They are “small” compared to proteins (di, tri, tetra… oligo-)

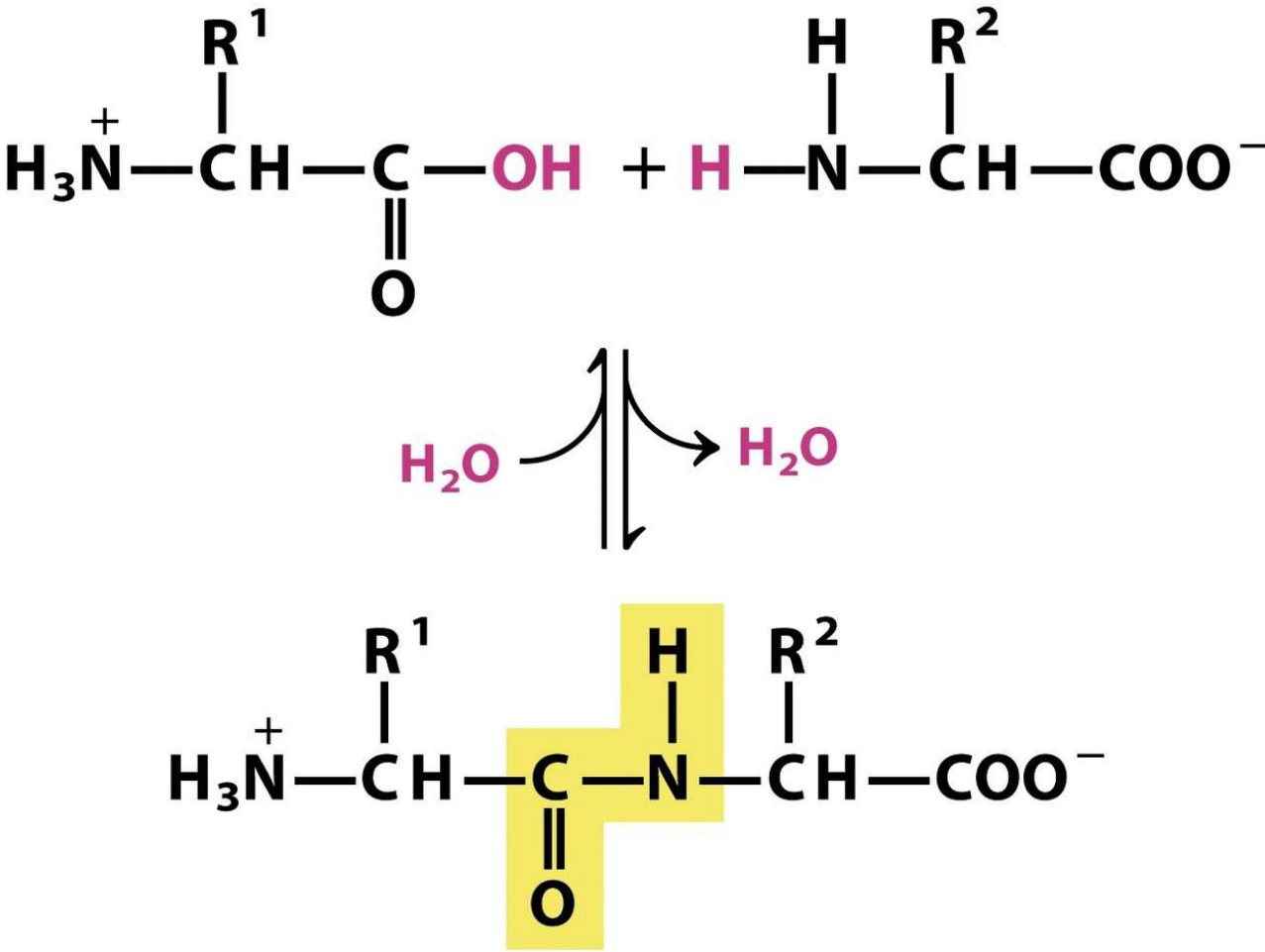


Figure 3-13
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Peptide Ends are Not the Same

Numbering starts from the amino terminus

AA_1 AA_2 AA_3 AA_4 AA_5

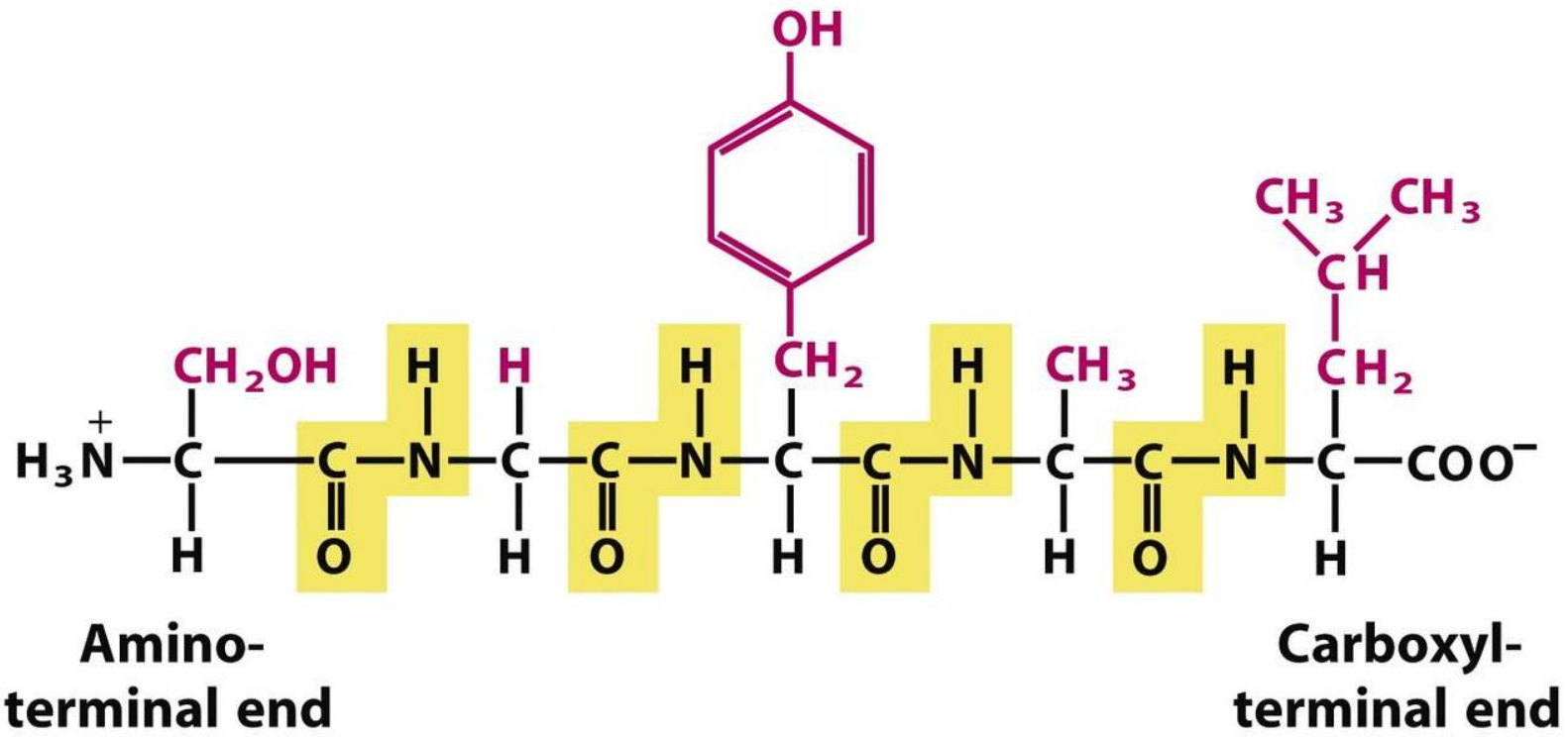


Figure 3-14
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The Three Letter Code

- Naming starts from the N-terminus
- Sequence is written as:

  Ala-Glu-Gly-Lys

- Sometimes the one-letter code is used:

  AEGK

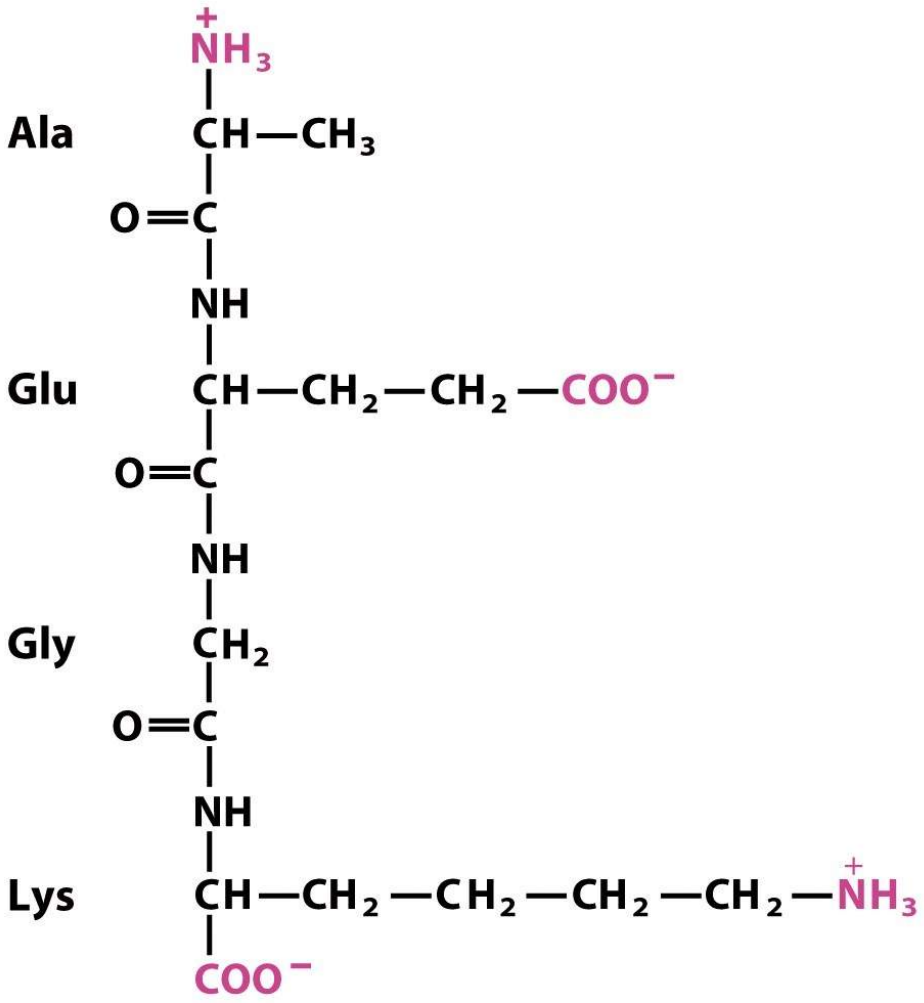


Figure 3-15
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